Electric Forces: Guided Notes

Chemistry · Unit 1B · Level 3 (closer to the class work) · goes with L1.7-L1.9 (Coulomb's Law, Neutral Interactions, Static Interaction Review)

Name: _____

Use the class slides and your reading. Key words:

electric force | *attract* | *repel* | *Coulomb's law* | *distance* | *amount of charge* | *scientific notation* | *exponent* | *polarized* | *neutral*

A. Attract or repel (L1.7, L1.9)

1. Opposite charges _____. Like charges _____.

2. Two balloons are both rubbed on hair. Predict what happens when they hang near each other, and explain.

_____

_____

B. Coulomb's law: proportional reasoning (L1.7)

3. More charge makes the force _____. More distance makes the force _____.

4. **Charge rule: multiply the force by the same number as the charge.**

> Example: The force is 6 mN. One charge doubles (× 2). New force = 6 × 2 = 12 mN.

a) Force is 5 mN. One charge triples (× 3). New force = _____

b) Force is 8 mN. One charge is cut in half (÷ 2). New force = _____

5. **Distance rule: square the change, then divide.**

> Example: The force is 36 mN. The distance doubles (× 2). 2 × 2 = 4, so new force = 36 ÷ 4 = 9 mN.
> If the distance gets smaller, multiply instead: half the distance → force × 4.

a) Force is 20 mN. Distance doubles. New force = _____

b) Force is 18 mN. Distance triples. New force = _____

c) Force is 3 mN. Distance is cut in half. New force = _____

C. Scientific notation (L1.7)

6. **Write the number out: move the decimal to the right as many places as the exponent.**

> Example: $2 \times 10^3 = 2{,}000$ $4.5 \times 10^4 = 45{,}000$

a) $6 \times 10^5 =$ _____

b) $2.3 \times 10^3 =$ _____

c) Which is more charge: 3×10^{12} e or 8×10^9 e? _____

D. Neutral objects (L1.8)

7. A neutral object near a charged object becomes _____.

8. Draw a negative balloon next to a neutral piece of paper. Show where the paper's electrons move. Draw an arrow for the force.

Electric Forces: Reading

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Name: _____

Pushes and Pulls Without Touching

Charged objects push or pull on each other without touching. This push or pull is called the electric force. Opposite charges (+ and −) attract. Like charges (− and −, or + and +) repel. Two balloons rubbed on your hair are both negative, so they push apart.

Charles-Augustin de Coulomb measured electric forces in the 1780s. His rule, Coulomb's law, says two things control the force: the amount of charge and the distance between the objects. More charge makes a stronger force. More distance makes a weaker force.

Distance matters a lot. If you double the distance, the force does not just get cut in half. It becomes one fourth (¼) as strong, because 2 × 2 = 4. If you triple the distance, the force becomes one ninth (1/9), because 3 × 3 = 9. Charge is simpler: double the charge on one object, and the force doubles.

The numbers of electrons are huge. A rubbed balloon can have about 2,000,000,000,000 extra electrons. Scientists write this in scientific notation: 2×10^{12}. The small number (the exponent) tells how many places the decimal moves. A bigger exponent means a much bigger number.

A charged object can even pull on a neutral object. When a negative balloon comes near paper, the paper's electrons are pushed to the far side. The near side becomes slightly positive. Since the positive side is closer to the balloon, the pull is stronger than the push, and the paper jumps up. We say the paper is polarized.

Check your reading

1. Name the two things that control the strength of the electric force.

2. Why does doubling the distance make the force ¼ as strong, not ½?

3. How can a balloon pull on paper that has no charge?

Electric Forces: Quick Check

Chemistry · Unit 1B · Level 3 (closer to the class work) · goes with L1.7-L1.9 (Coulomb's Law, Neutral Interactions, Static Interaction Review)

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Two objects with like charges will…

A. Attract | B. Become neutral | C. Repel

2. Which change makes the electric force stronger?

A. Adding more charge | B. Moving farther apart | C. Removing charge

3. The force between two charges is 16 mN. The distance doubles. The new force is…

A. 8 mN | B. 32 mN | C. 4 mN

4. 7×10^4 written out is…

A. 70,000 | B. 7,000 | C. 700,000

5. The force between two balloons is 10 mN. One balloon gets 3 times the charge. What is the new force? Show your thinking.

6. Explain why a charged balloon can pick up neutral paper. Use the words electrons and polarized.

Electric Forces: Guided Notes · ANSWER KEY

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electric force | *attract* | *repel* | *Coulomb's law* | *distance* | *amount of charge* | *scientific notation* | *exponent* | *polarized* | *neutral*

A. Attract or repel (L1.7, L1.9)

1. Opposite charges **attract**. Like charges **repel**.

2. Two balloons are both rubbed on hair. Predict what happens when they hang near each other, and explain.
They repel because both gained electrons and are negative (like charges).

B. Coulomb's law: proportional reasoning (L1.7)

3. More charge makes the force **stronger**. More distance makes the force **weaker**.

4. **Charge rule: multiply the force by the same number as the charge.**

> Example: The force is 6 mN. One charge doubles (× 2). New force = 6 × 2 = 12 mN.

a) Force is 5 mN. One charge triples (× 3). New force = **15 mN**

b) Force is 8 mN. One charge is cut in half (÷ 2). New force = **4 mN**

5. **Distance rule: square the change, then divide.**

> Example: The force is 36 mN. The distance doubles (× 2). 2 × 2 = 4, so new force = 36 ÷ 4 = 9 mN.
> If the distance gets smaller, multiply instead: half the distance → force × 4.

a) Force is 20 mN. Distance doubles. New force = **5 mN**

b) Force is 18 mN. Distance triples. New force = **2 mN**

c) Force is 3 mN. Distance is cut in half. New force = **12 mN**

C. Scientific notation (L1.7)

6. **Write the number out: move the decimal to the right as many places as the exponent.**

> Example: $2 \times 10^3 = 2{,}000$ $4.5 \times 10^4 = 45{,}000$

a) 6×10^5 = **600,000**

b) 2.3×10^3 = **2,300**

c) Which is more charge: 3×10^{12} e or 8×10^9 e? **3×10^{12} e (bigger exponent)**

D. Neutral objects (L1.8)

7. A neutral object near a charged object becomes **polarized** .

8. Draw a negative balloon next to a neutral piece of paper. Show where the paper's electrons move. Draw an arrow for the force.

Balloon with − signs; paper with − on the far side and + on the near side; arrow pulling the paper toward the balloon.

🔑 *Key written by our team from Mr. Brittle's worksheets and slides, please check.*

Electric Forces: Quick Check · ANSWER KEY

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Multiple choice and short answer. You may have it read aloud.

1. Two objects with like charges will…

A. Attract | B. Become neutral | **C. Repel**

2. Which change makes the electric force stronger?

A. Adding more charge | B. Moving farther apart | C. Removing charge

3. The force between two charges is 16 mN. The distance doubles. The new force is…

A. 8 mN | B. 32 mN | **C. 4 mN**

4. 7×10^4 written out is…

A. 70,000 | B. 7,000 | C. 700,000

5. The force between two balloons is 10 mN. One balloon gets 3 times the charge. What is the new force? Show your thinking.

30 mN (charge × 3, so force × 3).

6. Explain why a charged balloon can pick up neutral paper. Use the words electrons and polarized.

The balloon's charge pushes the paper's electrons to the far side; the paper is polarized, the near side is slightly + and is attracted.

🔑 *Reworded from his L1.9 review and L1.14 practice test, please check.*